

# MANTIK DEVRELERİ

Uize = %30  
Final = %50  
Lab & ödev = %20

Esma iBiş  
Ahmet METİN  
Sena Dikici  
lab  
hocaları

## Logic Circuits

## Sampling time

## quantization

### Definitions

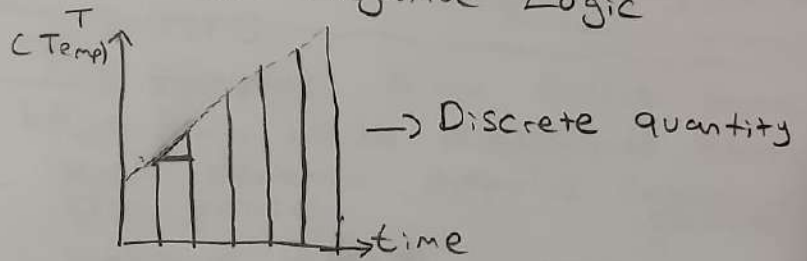
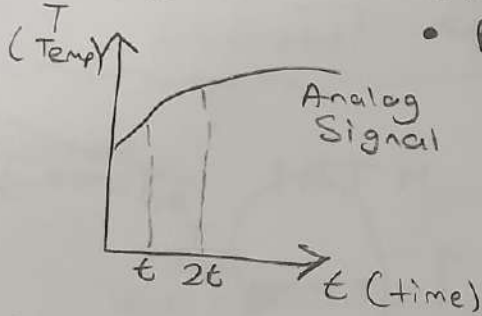
### Logic

### Gate

### Digital and Analog quantities

### Logic Levels

- Activation Level
- Positive and Negative Logic



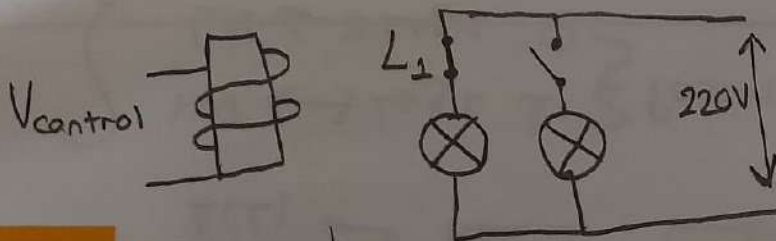
\* Zaman bakımından sürekli olan devrelere analog

\* Sinyal örneklenmiş, ayrılmış, kuantalanmış ise => digital devre

## Logic Levels

(H) High = 1 = Set = True = High Voltage (5V)

(L) Low = 0 = Reset = False = Low Voltage (0V)



$V_{control} = 0$   $L_1$ : on  $L_2$ : off

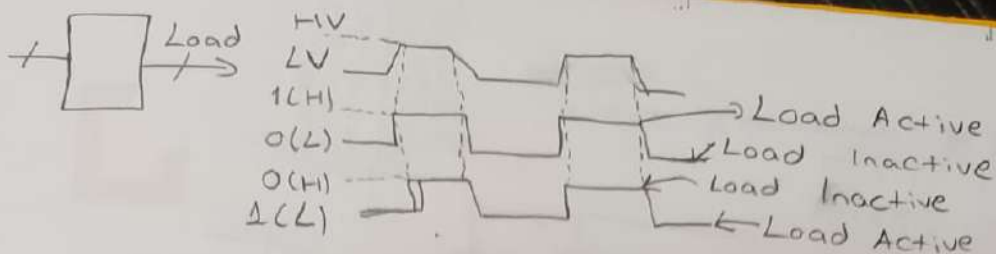
$V_{control} = 5V$   $L_1$ : off  $L_2$ : on

$L_1(L)$   $V_c = 0$

$L_2(H)$   $V_c = 5V$

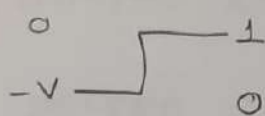
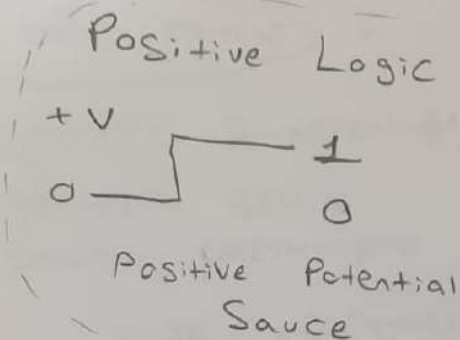
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| $L_2$ | $L_1$ | $L_0$ |       |
|-------|-------|-------|-------|
| 0     | 0     | 0     | 0     |
| 0     | 0     | 1     | 0,625 |
| 0     | 1     | 0     | 1,250 |
| 0     | 1     | 1     | 1,875 |
| 1     | 0     | 0     | 2,5   |
| 1     | 0     | 1     | 3,125 |
| 1     | 1     | 0     | 3,750 |
| 1     | 1     | 1     | 4,375 |



Logic Levels

|   |                              |   |                |
|---|------------------------------|---|----------------|
| } | HV $\leftrightarrow$ Logic 1 | } | Positive Logic |
|   | LV $\leftrightarrow$ Logic 0 |   |                |
|   | HV $\leftrightarrow$ Logic 0 | } | Negative Logic |
|   | LV $\leftrightarrow$ Logic 1 |   |                |



Negative Potential Source



Positive Potential Source

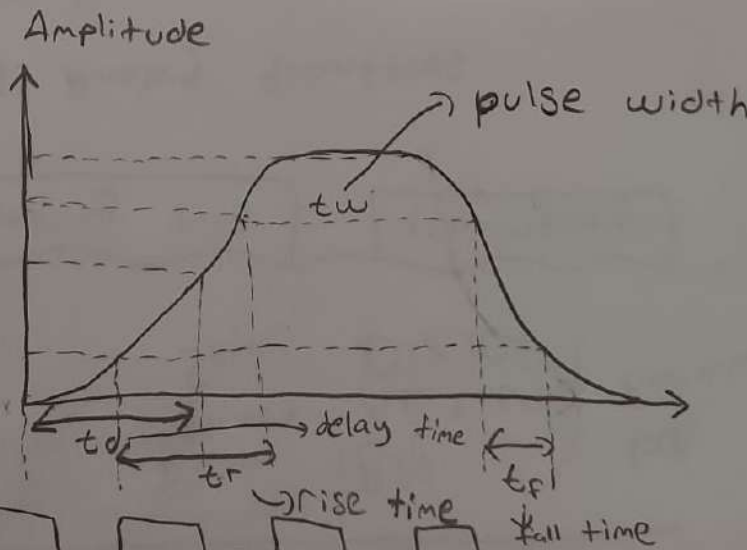
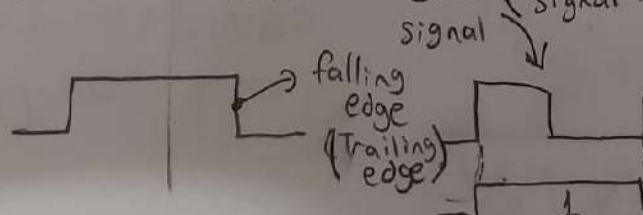
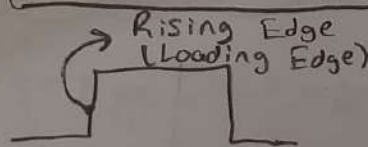
Example

H L L H

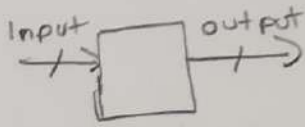
|    |                    |
|----|--------------------|
| PL | $\rightarrow$ 1001 |
| NL | $\rightarrow$ 0110 |

|           | TTL  | CMOS |
|-----------|------|------|
| HIGH      | 5V   | 5V   |
| UNCERTAIN | 2.0V | 3.5V |
| LOW       | 0.8V | 1.5V |
|           | 0    | 0    |

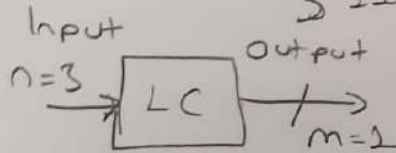
Pulse Waveform



## TRUTH TABLE



$2^n$



R (Red)

Y (Yellow)

G (Green)

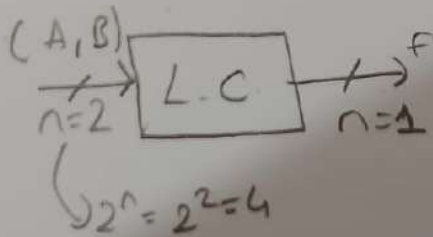
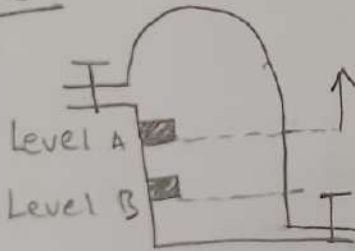
| Input   |   |   |   |
|---------|---|---|---|
| Decimal | A | B | f |
| 0       | 0 | 0 | 1 |
| 1       | 0 | 1 | 0 |
| 2       | 1 | 0 | 0 |
| 3       | 1 | 1 | 1 |

Example Bir trafik lambasındaki iki veya daha fazla farklı lamba aynı anda yandığında alarm veren devrenin doğruluk tablosunu çiziniz.

tablosu

| Dec | R | Y | G | F |
|-----|---|---|---|---|
| 0   | 0 | 0 | 0 | 0 |
| 1   | 0 | 0 | 1 | 0 |
| 2   | 0 | 1 | 0 | 0 |
| 3   | 0 | 1 | 1 | 1 |
| 4   | 1 | 0 | 0 | 0 |
| 5   | 1 | 0 | 1 | 1 |
| 6   | 1 | 1 | 0 | 1 |
| 7   | 1 | 1 | 1 | 1 |

## Example



Yağ Seviyesi A'nın çıktığı zaman veya B'nin altına düştüğünde alarm veren devrenin doğruluk tablosunu çiziniz.

| Dec | A | B | f |
|-----|---|---|---|
| 0   | 0 | 0 | 1 |
| 1   | 0 | 1 | 0 |
| 2   | 1 | 0 | 0 |
| 3   | 1 | 1 | 1 |

1. hafta

